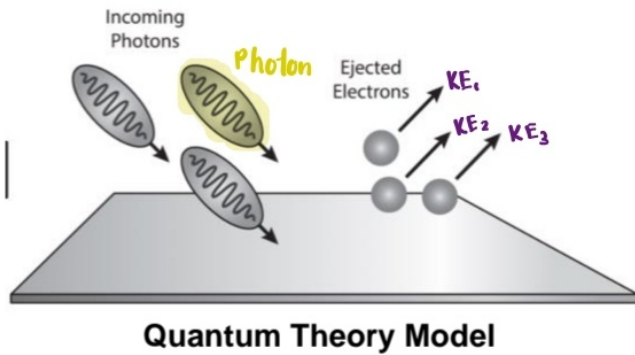


Photoelectric Effect



Working Principle

$$E_{\text{photon}} = hf_{\text{photon}} \rightarrow E_{\text{electron}}$$

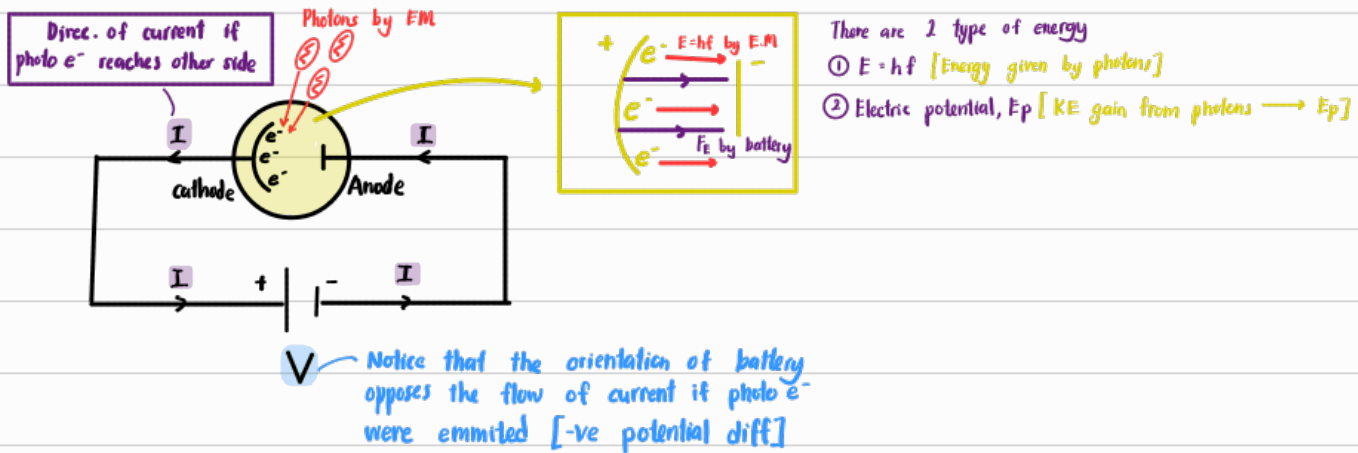
- ① When an e^- receives just enough energy, the e^- can jump out of its hole and reach the surface of the metal. [Energy to reach surface, E_{surface}]
- ② If more energy is given [$E > \phi$], that e^- will have excess KE and will be emitted from the metal surface. [$KE_{\text{emission}} = E_{\text{photon}} - E_{\text{surface}}$]
- ③ Since diff e^- are at diff depth in the metal, diff e^- have diff E_{surface} . However, the E_{surface} of the e^- closest to the surface [$E_{\text{surface minimum}}$] is called the work function, ϕ .
[$\phi = hf_{\text{surface minimum}} = hf_0$]
- ④ Thus, the maximum excess KE an e^- can have will be the e^- with the minimum E_{surface} .
[$KE_{\text{max emission}} = E_{\text{photon}} - \phi$]

Terminology

- ① Threshold f_0 : Minimum freq. the incident photon for photoelectrons to be emitted from metal surface. $E = hf$
Sub ↓
 $c = f\lambda$
- ② Threshold λ_0 : Maximum λ the incident photon for photoelectrons to be emitted from metal surface. $E = h\left(\frac{c}{\lambda}\right)$
- ③ Work function, ϕ : Minimum energy of incident photon for photoelectron to be emitted from metal surface

Stopping potential, V_s

- The negative potential diff across photocell to stop the photoelectron w the most K.E from reaching the other side.



Electrons might have diff work function because electrons can come from diff energy level
[Deeper the e^- , more energy needed to bring it to the surface]

$\therefore$ When $V \uparrow$, less & less photo e^- will be able to make it to the anode & will be attracted back to the positively charged plate [In this case, the cathode]
There will be a point where even the most energetic photo e^- can't overcome the -ve potential diff by the battery, causing it to be attracted back to the cathode, thus no current
That particular voltage is the stopping voltage, V_s

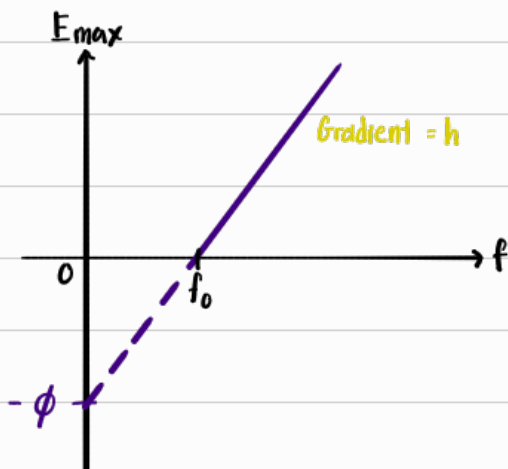
Graph of Einstein Eqn

$$KE_{\max} = E_{\text{photon}} - \phi$$

Extra energy after reaching the surface

$$E_{\text{net max}} = hf - hf_0$$

$$Y = mx + c$$



Since V_s is the voltage required to stop the photon w highest KE

$$\therefore V = \frac{E}{e}$$

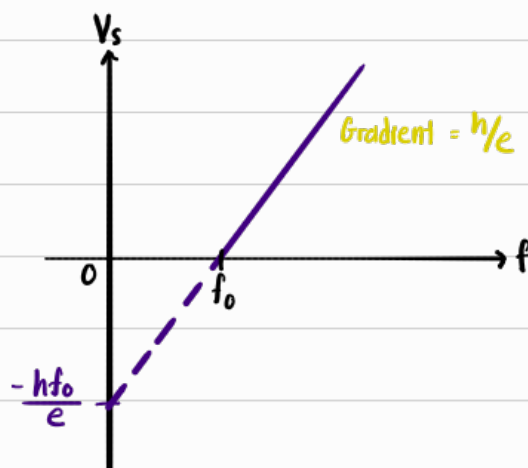
$$E = eV$$

$$E_{\max} = eV_{\text{stopping}}$$

$$\therefore E_{\text{net max}} = hf - hf_0$$

$$eV_s = hf - hf_0$$

$$V_s = \left(\frac{h}{e}\right)f - \left(\frac{h}{e}\right)f_0$$



Take note:

Since, $E_{\text{electron}} = eV$. Let V be 1V.

$$E_{\text{electron}} = (1.6 \times 10^{-19})(1) = 1.6 \times 10^{-19} \text{ J}$$

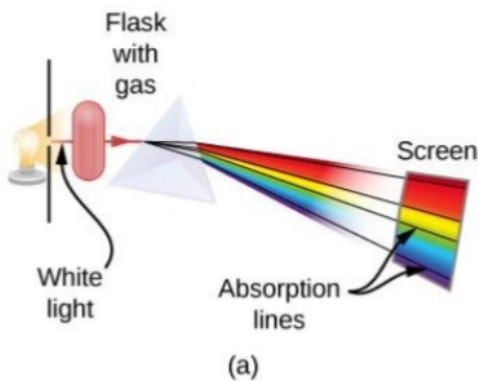
This value of $1.6 \times 10^{-19} \text{ J}$ is the KE gained by an e^- accelerating through a potential diff of 1V in a vacuum.

This value has been renamed to electronvolt, eV.

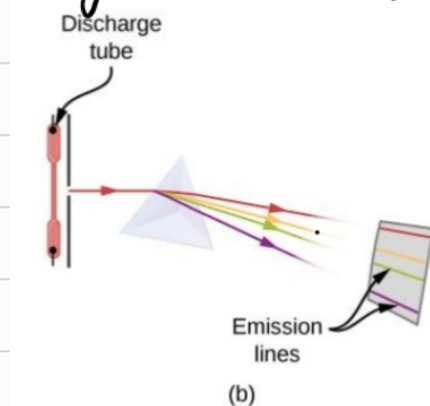
$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

Atomic spectra

Cold gas [Absorption spectrum]



Hot gas [Emission spectrum]



- 1) When light pass through,
 e^- absorb photons.
- 2) e^- move to higher energy lvl
- 3) $E_{\text{photon}} = \Delta E_{\text{level of } e^- \text{ in gas}}$
Since energy lvls are discrete,
only particular wavelengths are
absorbed.
- 4) When the e^- de-excites,
photons w those wavelengths
are released in all direction
[Not concentrated], creating
dark lines.

- 1) In hot gas, e^- are at higher
energy lvls
- 2) During de-excitation, e^- release
energy as photons w specific
wavelengths, corresponding to
coloured lines [Due to discrete energy lvls]

$\therefore$ Take Note:

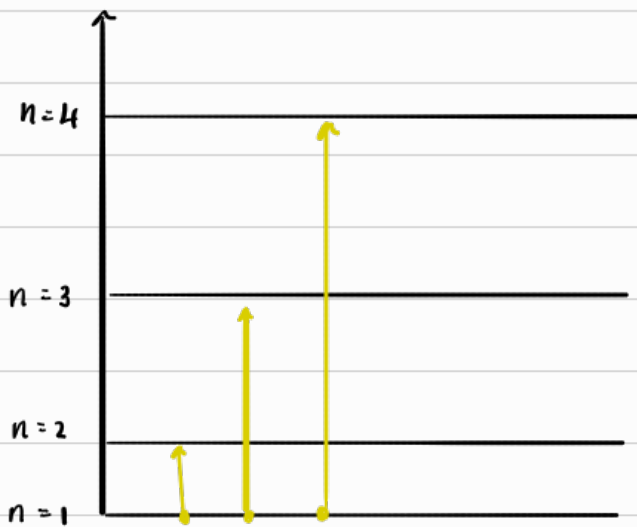
Absorption spectrum is a subset of the emission spectrum.

This is beuz for absorption spectrum, e^- normally start at $n=1$. [$n=1$ to $n=3$]
not $n=2$ to $n=3$, due to the temp of the gas.

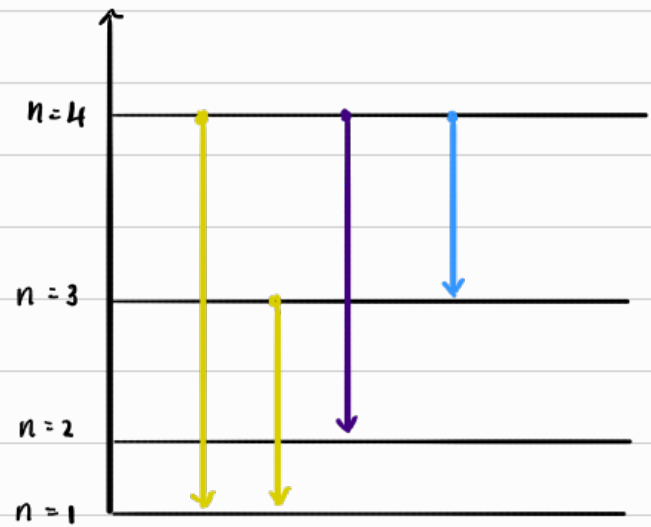
However for emission spectrum, e^- can start from any shell
[$n=4$ to $n=2$ or $n=3$ to $n=2$ etc].

Hydrogen spectral series

Absorption



Emission



↓ Lyman Series [Final lvl, $n_f = 1$]

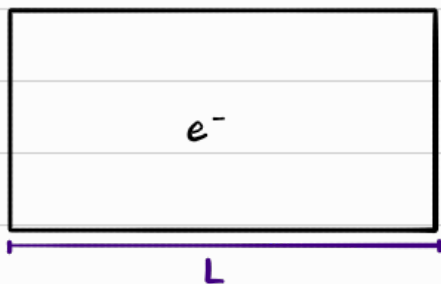
↓ Balmer Series [Final lvl, $n_f = 2$]

↓ Paschen Series [Final lvl, $n_f = 3$]

⋮

Particle in a box

Assume we have placed 1 particle [eg. $1 e^-$] inside a closed solid box, with no external forces, allowing the e^- to freely move [$PE = 0$]



At the quantum lvl, we can only predict where the e^- is but not know for sure until we physically observe it
[Open the box and see the e^-]

[Quantum lvl]

At particle lvl, Newton 2nd Law, $F=ma$ is not valid anymore.

We use Schrodinger Egn.

Wave function
[Contains info of where the particle might be]

$$E_n \psi_n(x) = -\frac{\hbar^2}{2m} \frac{d^2 \psi_n(x)}{dx^2} + u(x) \psi_n(x)$$

[Universal eqn]

Energy lvs
the e^- is
allowed to
have

[Allowed energies]

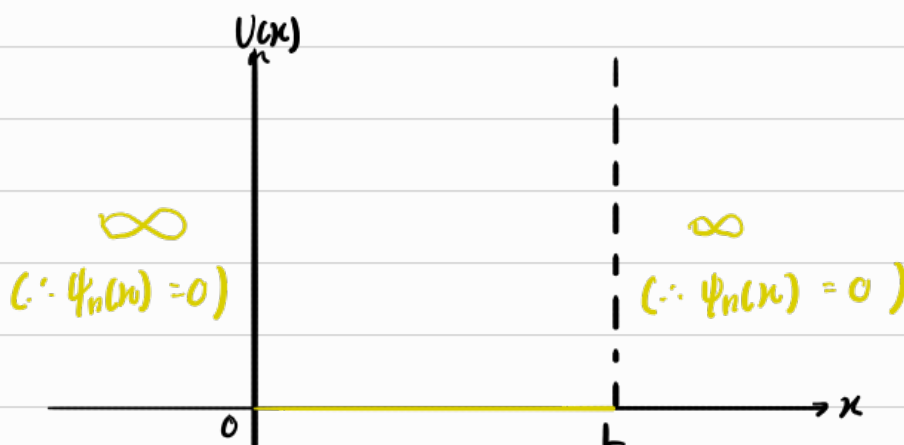
KE

PE

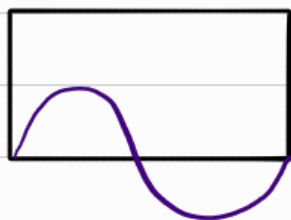
The solution to this differential equation gives us two solutions, E_n and $\psi_n(x)$.
This means for a particle w/ a particular energy lvl, E_n , then its wavefunction, $\psi_n(x)$ must behave in a certain way.

$x < 0$ or $x > L$

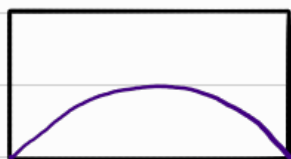
Since we know for sure that the particle won't be outside the box [$\psi_n(x) = 0$], we set $u(x)$ to be ∞ when $x < 0$ or $x > L$ [cuz when $u(x) = \infty$, the solution will be $\psi_n(x) = 0$]



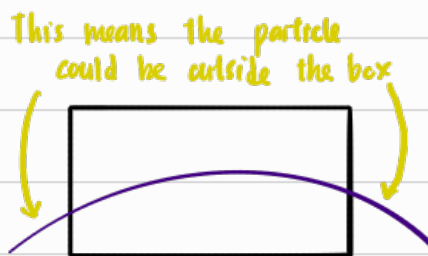
Since $\psi_n(x)$ must be zero outside the box, this means $\psi_n(x)$ must be zero at the corner of the box $[x=0 \text{ and } x=L]$. But if $\psi_n(x)$ is not zero at $x=0$ and $x=L$, means that $\psi_n(x)$ is non zero outside the box, that is impossible



Valid



Valid



Invalid

$\therefore$ Since $E = hf$, this means that an e^- only can be at a certain frequency inside the box.

$\therefore$ For "a particle in a box" scenario,

- $x < 0, x > L \rightarrow u(x) = \infty \rightarrow \psi(x) = 0$
- $0 < x < L \rightarrow u(x) = 0$

So anyhow, the term $u(x)\psi(x) = 0$.

$\therefore$ The schrodinger Egn simplifies to

$$E_n \psi_n(x) = -\frac{\hbar^2}{2m} \frac{d^2 \psi_n(x)}{dx^2}$$

Note that since $u(x) = 0$ inside the box,
 $TME = KE + PE$
 $E_n = KE$

The solutions to this differential eqn are,

$$\textcircled{1} E_n = \left[\frac{\hbar^2}{8mL^2} \right] n^2, \text{ where } n=1, 2, 3$$

$\Rightarrow E_n = (E_1) n^2$, where E_1 is the minimum energy

$$\textcircled{2} \psi_n(x) = \begin{cases} \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), & \text{where } n=2, 4, 6 \dots \text{ [Even numbers]} \\ \sqrt{\frac{2}{L}} \cos\left(\frac{n\pi x}{L}\right), & \text{where } n=1, 3, 5 \dots \text{ [Odd numbers]} \end{cases} \quad 0 < x < L$$

Hydrogen atom

The TME of an e^- in a hydrogen atom at a certain energy lvl is

$$TME = (-13.6 \text{ eV}) \left(\frac{1}{n^2} \right)$$

$$E_n = (-13.6 \text{ eV}) \left(\frac{1}{n^2} \right)$$

Allowed energy lvl's

Hydrogen-like atoms

Atoms w same no. of e^- , but diff no. of protons, z .

Example : $\text{He}^+ \rightarrow 1e^- + 2 \text{ protons} \quad [z = 2]$

$\text{Li}^{++} \rightarrow 1e^- + 3 \text{ protons} \quad [z = 3]$

$$E_n = (-13.6 \text{ eV}) \left(\frac{z^2}{n^2} \right)$$